

Hardaat Singh Baath

✉ hardaatsinghbaath@gmail.com

☎ +91 765 785 2262  LinkedIn

 Website  GitHub

EDUCATION

B.E. in Computer Science, *Birla Institute of Technology and Science, Pilani* 

November 2021 – present
Sancoale, Goa, India
– CGPA: 7.45

SKILLS

Programming Languages

C, C++, Python, Java, NodeJS, JavaScript, MATLAB

Libraries

Numpy, Matplotlib, Pandas, Seaborn, OpenCV, Scikit-learn

Framework

ROS, PyTorch, Tensorflow, Keras

Tools

Gazebo, CoppeliaSim, Docker, AWS, Git and GitHub, Arduino, Google Sheets

Soft Skills

Public Speaking, Canva, Creative Writing

Languages

English, Hindi, Punjabi

RELEVANT COURSES

Computer Programming | Data Structures and Algorithms | Database Management Systems | Operating Systems | Computer Networks | Computer Architecture | Probability and Statistics | Linear Algebra | Differential Equations | Multivariate Calculus | Machine Learning | Brain Inspired Deep Learning

Online Courses

- AMRx: Autonomous Mobile Robotics, ETH Zurich 
- CS231n Stanford Computer Vision 
- IBM AI Engineering Specialization 

AWARDS

National Qualifiers, NXP AIM 2024

August 2024

- Among top 15 teams that took part in NXP AIM 2024 Regionals.

Excellence Award (Overall),

International Rover Challenge, Bangalore
January 2023

- Won Best Overall Rover award, 2nd in the Autonomous Challenge and 4th overall.

EXPERIENCE

AI Engineer, DG Takano

May 2024 – July 2024 | Japan

- Developed an automated testing system with **ESP32** for **Cloud Data logging**, improving productivity, speed and reducing manual effort
- Integrated **Arduino Cloud** and **Slackbot** for real time alerts, enabling live remote monitoring of the setup.
- Developed an **Automated G-code generation system** that accelerated prototyping by learning from previous data, significantly improving the production of G-Code generation for machine operators.

Research Intern, CSIR - CEERI, Pilani

June 2023 – August 2023 | Pilani, Rajasthan, India

- Advisor: Dr Dhiraj Sangwan 
- Developed a DL-based pipeline for damage generation, segmentation and restoration of Rajasthani Wall Murals.
- Used models like **StyleGAN2-ada-PyTorch** for damage generation, **UNet++** and **DeepLabV3+** for damage segmentation and **AOT-GAN** and **Partial Convolutional Network** for image inpainting.

PUBLICATIONS

Damage Segmentation and Restoration of Ancient Wall Paintings for Preserving Cultural Heritage,

International Conference on Computer Vision and Image Processing, 2023 

Hardaat Singh Baath, Soham Shinde, Jinam Keniya, Priyanshu Ranjan Mishra, Anil Saini, Dhiraj Sangwan

PROJECTS

Transformers for Stock Market Prediction, Supervised Project (ongoing)

February 2024 – present

- Advisor: Dr. J K Sahoo 
- Exploring and implementing an **Transformers-based time series algorithm** for Stock Market Prediction using features like **Signal line**, **MacD line**, **EMA** etc with **CNN-based** feature extraction.
- Implementing a trading algorithm with transaction costs and taxation for realistic simulation of stock markets using **NIFTY 50** and **S&F 500** market datasets.

NeuroChaos Learning with Chaotic Maps, Supervised Project (ongoing)

August 2024 – present

- Advisor: Dr Hari Krishnan N B 
- Exploring the effect of various chaotic maps like Skew-tent, Gauss, Circle Maps etc for **feature extraction** for ML Applications.
- Analysing the **effect of these chaotic features** on the speed and accuracy of various ML models on standard datasets like iris, mnist, etc.
- Working on designing a **novel architecture** that leverages chaotic features for enhanced model performance.

Distributed Deployment Platform, Course Project

January 2024 – June 2024

- Implemented a distributed deployment platform that deploys Github repositories on web using **AWS S3** and **ECS** services.
- Added backend with **API server** and **Reverse Proxy** using **NodeJS** for automated deployment and real time logs..
- Logs saved on **PostgreSQL Database**, using **Kafka** Server as an intermediate service.
- Added support for different types of languages like ReactJS, NodeJS, HTML etc to enable **language based deployment**.

Project Kratos, A Mars Rover, Club Project

May 2023 – May 2024

- Worked on a **P-controlled visual servo** algorithm to follow Arrow and ArUco tags detected using **YOLOv3**, **OpenCV** and **Python**.
- Implemented **RTK-GNSS-based GPS** coordinates system and **PID-based** algorithm to navigate to the specified points.
- Implementing **Probabilistic Terrain Mapping** algorithm and global and local planner using Jetson Xavier AGX and Zed 2i

MENTORSHIP

Subsystem Lead, Autonomous Subsystem, Project Kratos

May 2023 – May 2024 | Sancoale, India

- Leading a team of 10+ members in the Autonomous System.
- Doing Research and Development of the current system and involved in collaboration with other subsystems